

BENCHMARK: NON-DETECTION

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2455792.8853 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.149 +/- 0.06                  |
| Transit depth (Rp/Rs)^2                           | 2.23 +/- 1.79 [%]               |
| Orbital Inclination (inc)                         | 84.28 +/- 1.86                  |
| Airmass coefficient 1 (a1)                        | 0.5 +/- 0.31                    |
| Airmass coefficient 2 (a2)                        | 0.74 +/- 1.57                   |
| Scatter in the residuals of the lightcurve fit is | 61.35 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 3.0                             |
| Optimal Annulus                                   | 5.5                             |
| Transit Duration (day)                            | 0.031 +/- 0.033                 |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.149 ± 0.06 vs 0.1592 (deviation 0.11σ) |
| Duration fit vs published | 0.031 d vs 0.0946 d (deviation 1.3σ)     |
| Mid-time O–C              | 1.9 min                                  |
| Depth SNR                 | 1.7                                      |
| Residual scatter          | 61.35 %                                  |

Light curve

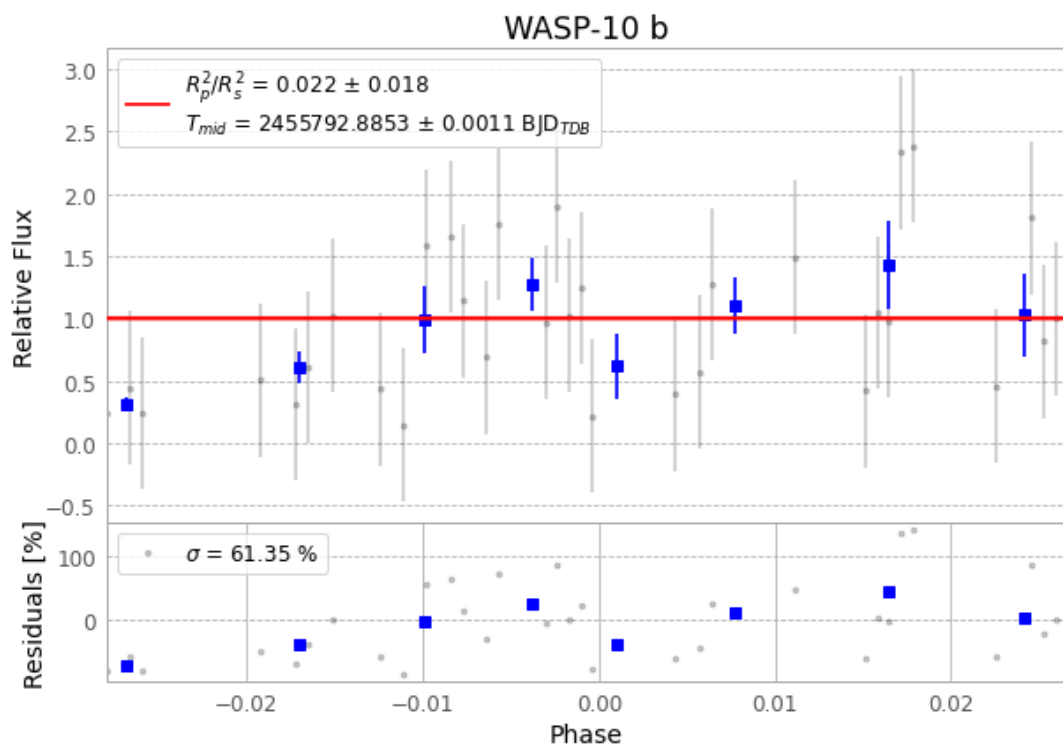


Figure 1 — FinalLightCurve\_WASP-10 b\_2011-08-19.png (SHA-256 864b09186988f6d5...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 321], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8baa9e10f910b5e2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

85 FITS frames (56 MB), WASP-10110819070312.FITS → WASP-10110819111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-110819.log (446df98c3e04...), FinalLightCurve\_WASP-10 b\_2011-08-19.png (864b09186988...), FinalLightCurve\_WASP-10 b\_2011-08-19.csv (7412e483fb5f...)

## Compute resources

Wall time 133.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:54Z.